

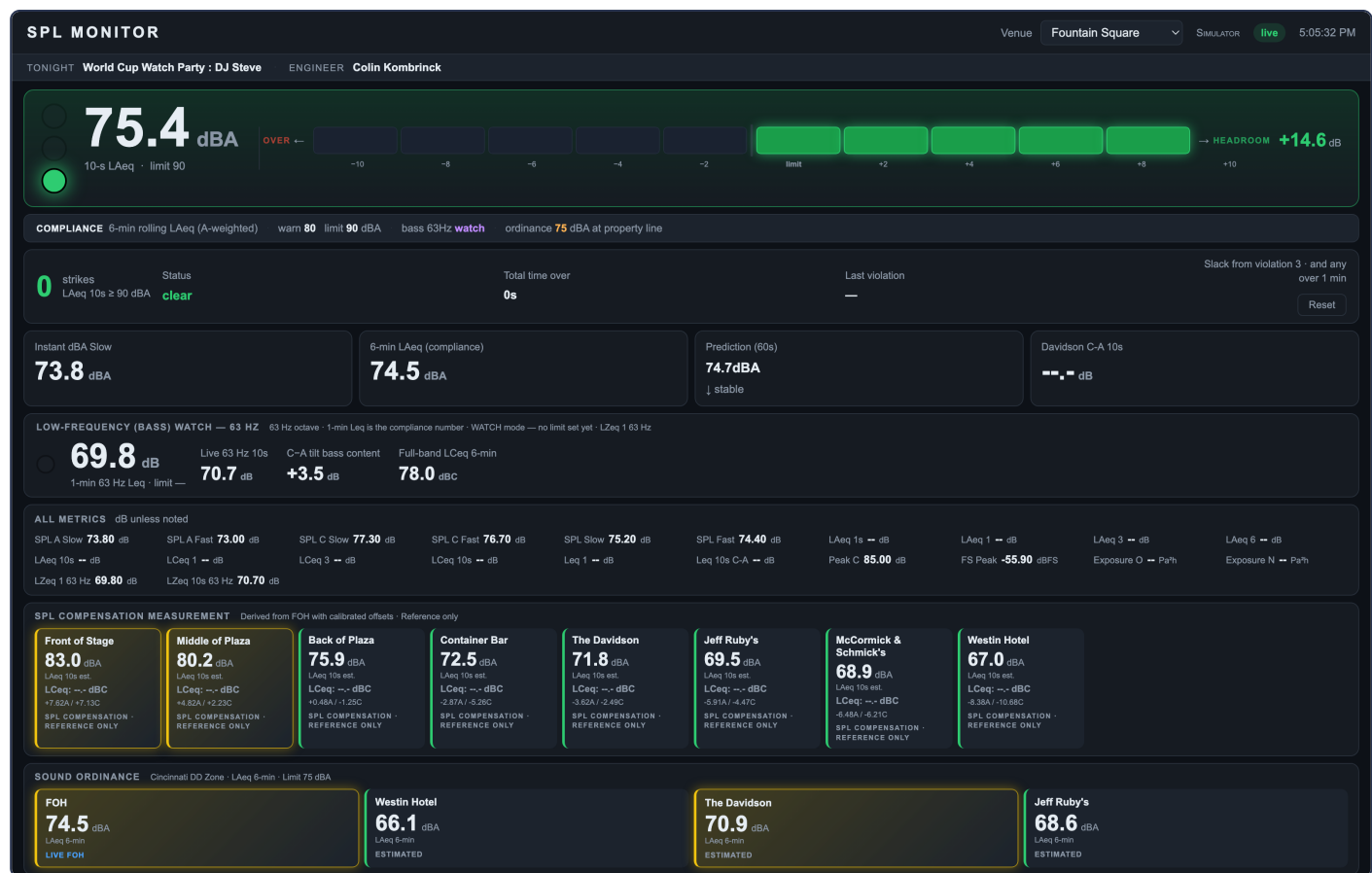
Reading the SPL Dashboard

A plain-language walkthrough of what's on screen, where the numbers come from, and how the alert system works — built for anyone who doesn't spend their nights staring at a sound level meter.

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Prepared by Brian Lloyd · Tiny Door Studios · Screens below are the live dashboard running in Simulator mode with a test signal, so the numbers are safe to look at but not from a real show.

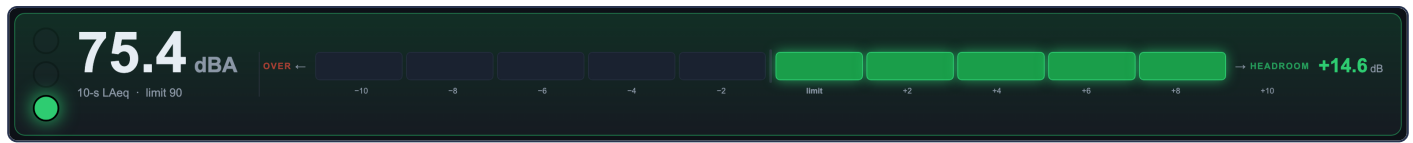
Dashboard Overview, At a Glance



Full dashboard view, spl.tinydoorstudios.com — Simulator mode, Fountain Square

Here's the whole screen before we zoom into pieces of it. Top bar: venue selector, a tag showing **Simulator** or **live**, and a clock — always check that tag first so you know whether you're looking at a real show. Below that, top to bottom: the hero reading and headroom bar (Section 1); the compliance strip and strike tracker (Sections 3–5); four quick tiles for instant level, 6-minute compliance, a short-term prediction, and one estimated location; the Low-Frequency Bass Watch panel; a full metrics grid for anyone who wants every number Smaart is tracking; then the SPL Compensation row and Sound Ordinance panel (Sections 2–3). Everything on screen updates live, and the whole layout fits without scrolling.

1 The Hero Number



Live screen capture, spl.tinydoorstudios.com — Simulator mode, Fountain Square

The big number top-left is the **current reading** — a 10-second rolling average (labeled **10-s LAeq**) from our one calibrated measurement microphone at front-of-house. Think of it as a speedometer: it's telling you what's happening right now, smoothed just enough that it isn't jumping around on every kick drum hit.

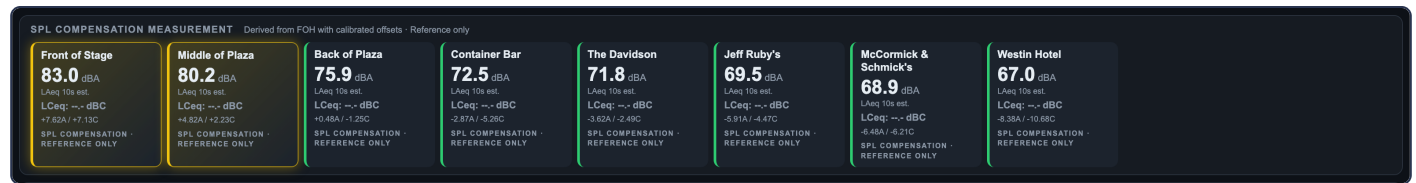
Next to it is the **limit** — our internal working ceiling for the show (90 dBA in this example). The row of segments is a headroom gauge, centered on that limit line: it fills green to the right as long as we're under the limit, and more lit segments means more buffer before we'd cross it. If we went over, segments on the **"OVER"** side (left) would light up instead, showing how far past the limit we'd gone. **Headroom**, top right, is that same buffer expressed as one number — the gap left before we'd hit the limit.

That's the default. When the limit switch is set to the 95 dBA rule (Section 5), the big number changes to an instant **SPL A Slow** reading, the label under it changes to match, and the limit reads 95.

Why a 10-second average. A single instant reading bounces all over the place and isn't useful to make decisions from. Ten seconds is short enough to react to in real time, long enough to actually mean something.

spl.tinydoorstudios.com — accessible from any laptop, tablet, or phone with an internet connection, no software install required.

2 Virtual Readings — and Why They're Trustworthy



SPL Compensation Measurement row — Front of Stage is the one physical mic; everything else is calculated

We only have one physical, calibrated measurement microphone out there, positioned at front-of-house. Every other card on that row — Middle of Plaza, Back of Plaza, Container Bar, The Davidson, Jeff Ruby's, McCormick & Schmick's, Westin Hotel — is a **virtual reading**. There's no second microphone sitting on the patio at Jeff Ruby's. Those numbers are calculated.

Here's how that calculation stays honest: our measurement software (Smaart) supports a feature called **SPL Compensation**. A reference signal gets measured at both the FOH mic position and each of those other spots, and the software works out a fixed dB offset between them. From then on, that offset rides along with the live FOH reading, so each card shows what a microphone placed at that spot actually would read — without needing a microphone actually placed there.

That's also why every card is labeled "**SPL Compensation · Reference Only**" right on the dashboard itself. That's not a caveat we added for this guide — it's built into the software: calibrated estimates anchored to one real physical measurement, not a second mic quietly sitting at each spot.

Note: this offset is set during a calibration pass against a physical reference standard — it isn't something we redo before every single show.

3 Two Numbers That Matter: 10-Second vs. 6-Minute

COMPLIANCE 6-min rolling LAeq (A-weighted) warn **80** limit **90** dBA bass 63Hz **watch** ordinance **75** dBA at property line

The compliance strip — our working limit and the city's separate ordinance number, side by side

The dashboard tracks two different limits, and they're easy to mix up:

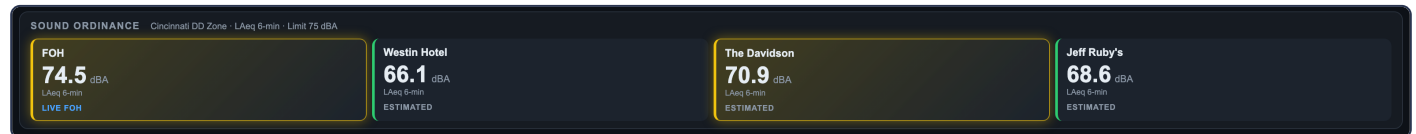
Our working limit

90 dBA, measured as a **10-second average at FOH**. This is the number we watch to make mix decisions and the number that drives the strike system (more on that next page). It's ours — a mixing guardrail, not a legal one.

City ordinance

75 dBA, measured as a **6-minute rolling average at the property line**. This is Cincinnati's actual Downtown Development Zone sound ordinance number — the one that matters legally, and it's measured out at the edge of the site, not at our mix position.

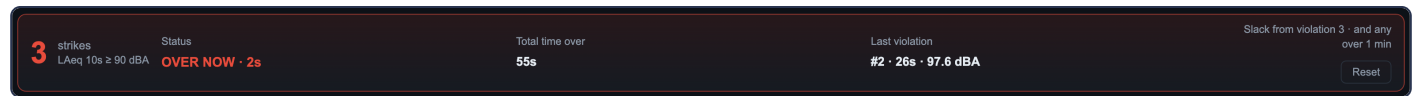
A loud moment at FOH can spike the 10-second number without ever pushing the 6-minute property-line average anywhere near the ordinance limit — that's normal. The dashboard estimates that property-line number continuously so we're never guessing:



Sound Ordinance panel — FOH is a live measurement; the surrounding property lines are calculated estimates (see previous page)

FOH is tagged "**Live FOH**" because it's the real, direct measurement. The property-line figures (Westin Hotel, The Davidson, Jeff Ruby's) are tagged "**Estimated**" because they're calculated the same way the virtual readings on the previous page are — offset from the one physical mic, not a second mic on someone else's roof.

4 The Three-Strike System



Strike tracking panel — live counter, current status, total time over limit, and the most recent violation

Every time the 10-second FOH reading crosses our working limit (90 dBA under the default rule; see Section 5 for the 95 A-Slow rule), it's logged as a **strike**, tracked toward a maximum of three. The panel above catches it mid-violation: **3 strikes** logged, status showing **OVER NOW** — we're above the limit at this exact moment, two seconds into strike three — with a running total of **55 seconds** spent over the limit so far tonight.

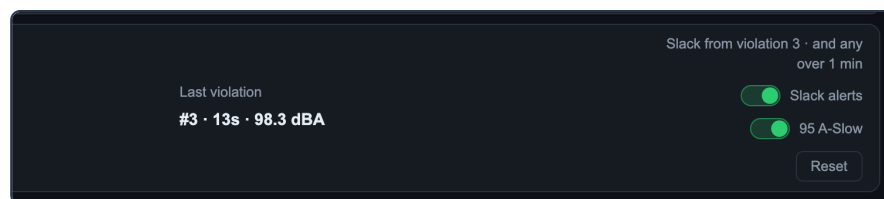
Duration is tracked per incident, not just counted. **Last violation: #2 · 26s · 97.6 dBA** is the last *completed* strike — number three is still running, so the most recent one on the books is number two, which lasted 26 seconds and peaked at 97.6 dBA. That's what lets us have a real conversation afterward instead of a vague one: we know not just that it happened, but how long it ran and how hot it got.

The Slack alert. Any strike of 3 or higher automatically posts to our internal Slack channel in real time, with the violation number, duration, and peak level included — the same detail shown on screen. Nobody has to be staring at this dashboard for a violation to get flagged and logged.

The **Reset** button clears the strike counter between shows — it doesn't erase the historical log, it just starts the next event's count at zero.

Reset is PIN-locked to Brian. Clearing the counter takes a PIN code that only Brian holds — engineers can't reset it. That's on purpose: it means nobody working a show can wipe a night's strikes to hide that they went over the limit. The count on screen is the count that stands.

5 The Limit Rule Switch: 90 LAeq 10s or 95 dBA Slow



Strike panel, right side: the Slack alerts switch and, below it, the limit rule switch, shown here set to 95 A-Slow

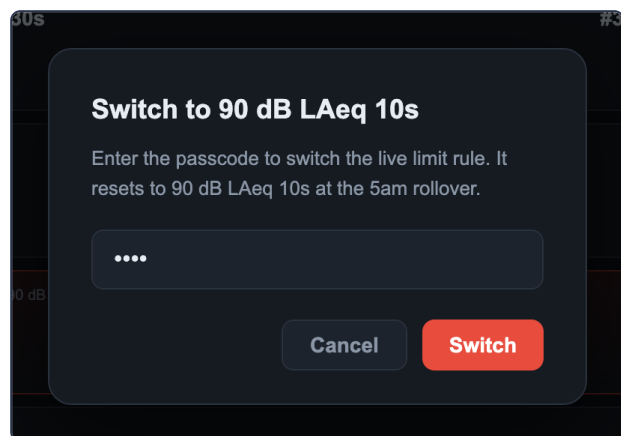
Some shows run under a different house rule: instead of a 10-second average capped at 90, the cap is **95 dBA on an instant A-weighted Slow reading**. The switch under Slack alerts picks which rule the dashboard enforces. Off (grey) reads **90 LAeq10s**, our standard. On (green) reads **95 A-Slow**.

90 LAeq10s (default)

Hero number, traffic light and headroom all run off the 10-second average. Limit 90, yellow from 80. Any reading at or over 90 is a strike right away — the 10-second averaging already smooths out single hits.

95 A-Slow

Everything switches to the instant SPL A Slow reading. Limit 95, yellow from 90. A strike needs **3 seconds** straight at or over 95, so one loud kick or cymbal crash won't count.



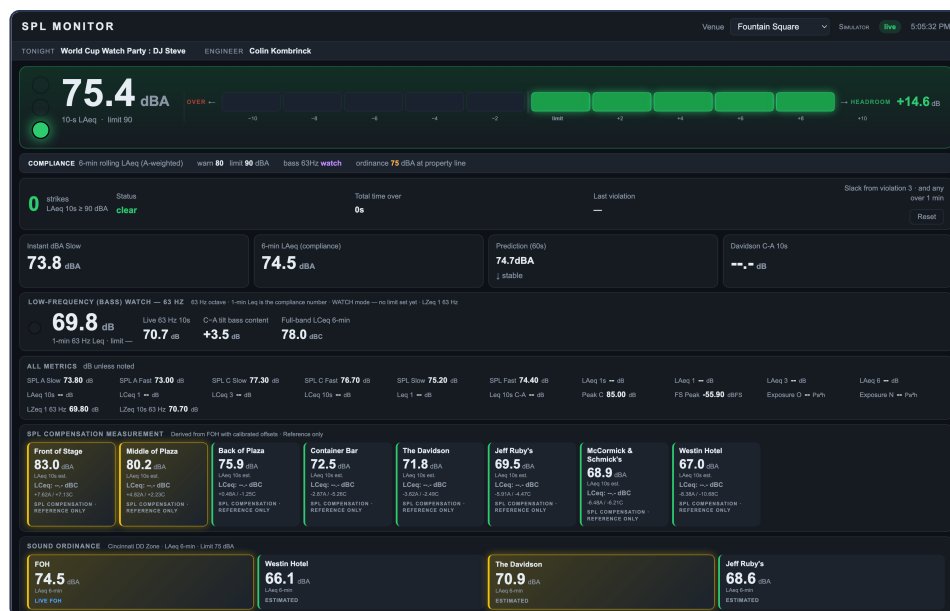
Changing the rule asks for the PIN, in both directions

Flipping the switch **either way** asks for the same PIN as Reset, so like Reset it's Brian's call, not the engineer's. The switch doesn't move until the PIN is accepted, and every screen watching the dashboard updates at the same moment. Strikes already on the board stay put; if a violation is in progress when you switch, it's closed out and recorded under the rule it started on.

It resets itself. At 5am each day the dashboard goes back to 90 LAeq10s on its own, so a 95 night never carries into the next show. If the system restarts mid-show, it keeps whatever rule was set.

The nightly report records which rule was in force and when it changed, and the graphs draw the limit line for each rule used that night.

6 Where the Data Comes From, and How to Check It Yourself



Full dashboard view — every panel covered in this guide, together in context

- 1 **Smaart** (our field-standard SPL logging software) reads the FOH measurement mic continuously through the show. When it's a real show, the dashboard shows the actual input source (a DiGiCo console feed); in these sample screens it reads **"Simulator"** instead, so you can tell at a glance that this data isn't from a live event.
- 2 Smaart streams that live data out continuously, which is what powers the dashboard at **spl.tinydoorstudios.com** in real time — the page you're looking at updates the whole time doors are open.
- 3 Every night, Smaart separately **generates and emails a full SPL log report** — an independent record of the night, separate from the live dashboard feed, so we've still got a log if the connection ever drops mid-show.

The live dashboard and the report that lands in the inbox the next morning trace back to the same measurement chain — just two different snapshots of it.

Access. The dashboard is browser-based at **spl.tinydoorstudios.com** — it opens on any laptop, tablet, or phone with an internet connection, no software install needed. The venue selector at top lets us switch between monitored sites as more come online.

Just another tool — not another task. This runs completely on its own. As long as you're logging SPL for your show the way you already do, everything here just works — the live screen, the strike tracking, the alerts, and the nightly report all happen automatically. There's nothing extra to set up, open, or babysit. It's here to make the night easier, not to add to your plate.

A living dashboard. This is an evolving web page — panels and features will change and grow over time. We'll do our best to keep this guide in step with the dashboard as those changes land.